Data Analytics using Python

Week 8 Question Bank Answer

1 Explain Maximum Likelihood Estimation (MLE).

Ans.1) Maximum Likelihood Estimation (MLE) is a method used to estimate the parameters of a statistical model. The key idea behind MLE is to find the parameter values that maximize the likelihood of observing the given data.

The steps involved in MLE are:

1. Assume a probability distribution for the data, which depends on one or more unknown parameters.

2. Write the likelihood function, which is the joint probability of observing the given data under the assumed distribution and the unknown parameters.

3. Find the parameter values that maximize the likelihood function. This is typically done by taking the derivative of the log-likelihood function and setting it equal to zero.

4. The values of the parameters that maximize the likelihood function are called the maximum likelihood estimates (MLEs).

The main advantages of MLE are:

- The resulting estimators have desirable statistical properties, such as being asymptotically unbiased, consistent, and efficient.

- MLE can be applied to a wide range of probability distributions, not just the normal distribution.

- MLE provides a principled way to estimate parameters by maximizing the agreement between the data and the model.

In summary, MLE is a powerful technique for estimating unknown parameters in a statistical model by finding the values that make the observed data most likely. It is widely used across many fields of science and statistics.

2 What do you mean by logistic regression and explain when it is used?

Ans.2) Logistic regression is a statistical modeling technique used to predict the probability of a binary or categorical outcome variable based on one or more predictor variables. It is particularly useful when the dependent variable is dichotomous, meaning it can only take two possible values, such as "yes/no", "success/failure", or "approved/rejected".

The logistic regression model has the form:

$p = 1 / (1 + e^{(-z)})$

The key aspects of logistic regression are:

1. **Dependent variable:** The dependent variable in logistic regression is binary or categorical, typically coded as 0 or 1, where 0 represents the absence of an event and 1 represents the presence of an event.

2. **Probability modeling:** Logistic regression models the probability of the dependent variable being 1 (the event occurring) as a function of the predictor variables. The model estimates the probability of the event occurring, which ranges between 0 and 1.

3. **Logit transformation:** Logistic regression uses the logit transformation to model the relationship between the predictor variables and the probability of the event occurring. The logit is the natural logarithm of the odds ratio, which is the ratio of the probability of the event occurring to the probability of the event not occurring.

Logistic regression is used in a variety of applications, including:

- Predicting the likelihood of a customer making a purchase (e.g., in e-commerce)
- Determining the risk of a patient developing a certain disease (e.g., in medical diagnosis)
- Classifying email messages as spam or not spam (e.g., in email filtering)
- Estimating the probability of a loan applicant defaulting on a loan (e.g., in credit risk assessment)
- Predicting the outcome of a political election (e.g., in political science research)

The key advantage of logistic regression is its ability to handle binary or categorical dependent variables, which are common in many real-world applications. It provides a straightforward way to estimate the probability of an event occurring and to understand the relationship between the predictor variables and the outcome.

3 What is G test statistic used in logistic regression?

Ans.3) The G test statistic, also known as the likelihood ratio test, is a statistical test used in logistic regression to assess the overall fit of the model and to compare the fit of different models.

In logistic regression, the G test statistic is used to test the null hypothesis that all the regression coefficients (except the intercept) are equal to zero. This is equivalent to testing whether the model with the predictors provides a significantly better fit to the data than the null model (the model with only the intercept).The formula for the G test statistic is:

G = -2 * ln(L0/L1)

Where:

- L0 is the likelihood of the null model (the model with only the intercept)
- L1 is the likelihood of the full model (the model with the predictors)

The G test statistic follows a chi-square distribution with degrees of freedom equal to the number of predictors in the full model (excluding the intercept).The steps to perform the G test in logistic regression are:

1. Fit the null model (the model with only the intercept) and obtain the log-likelihood (L0).
2. Fit the full model (the model with the predictors) and obtain the log-likelihood (L1).
3. Calculate the G test statistic using the formula: G = -2 * ln(L0/L1).
4. Determine the p-value by comparing the G test statistic to a chi-square distribution with degrees of freedom equal to the number of predictors in the full model (excluding the intercept).

If the p-value is less than the chosen significance level (e.g., 0.05), you can conclude that the full model provides a significantly better fit to the data than the null model, and at least one of the predictors is significantly associated with the outcome variable.The G test is a useful tool in logistic regression to assess the overall model fit and to compare the fit of different models, which can help in selecting the most appropriate model for the data.

4 Compare Linear Regression model and Logistic regression model.

Ans.4)

Aspect	Linear Regression	Logistic Regression
Dependent Variable	Continuous	Binary or Categorical (Probability)
Output	Continuous values	Probabilities (0 to 1)
Model Form	Linear relationship between variables	Logit transformation to model log-odds of the dependent variable
Assumptions	Linearity, normality of residuals, constant variance	Linearity of log-odds, independence of errors, no multicollinearity
Evaluation Metrics	R-squared, Mean Squared Error (MSE), Root Mean Squared Error (RMSE)	Accuracy, Precision, Recall, Area Under the ROC Curve (AUC-ROC)
Application	Predicting continuous outcomes like sales, temperature, stock prices	Predicting binary outcomes like yes/no, spam/not spam, credit risk assessment
Example Use Case	Predicting house prices based on area, number of bedrooms, location	Predicting customer churn based on demographics, predicting loan default based on credit score
Parameter Estimation	Least Squares method to minimize residuals	Maximum Likelihood Estimation to maximize likelihood of observing the data
Model Fit	Measures how well the model fits the data using R-squared and error metrics	Measures how well the model predicts probabilities using metrics like accuracy, AUC-ROC

This table summarizes the key differences between Linear Regression and Logistic Regression models in terms of their dependent variable, output, model form, assumptions, evaluation metrics, applications, example use cases, parameter estimation methods, and model fit evaluation.

OR

Linear regression and logistic regression are two commonly used statistical methods for modeling relationships between variables. The main differences between the two methods are the types of dependent variables they can handle and the underlying assumptions about the relationship between the variables.

Linear regression is used when the dependent variable is continuous, and the relationship between the dependent and independent variables is assumed to be linear. The goal of linear regression is to find the line of best fit that minimizes the sum of the squared residuals, which are the differences between the observed and predicted values of the dependent variable.

Logistic regression, on the other hand, is used when the dependent variable is binary or categorical, and the relationship between the dependent and independent variables is assumed to be nonlinear. The goal of logistic regression is to find the curve of best fit that maximizes the likelihood of observing the data given the model parameters. The curve is typically an S-shaped curve called the logistic function, which maps the linear combination of the independent variables to a probability between 0 and 1.

Another key difference between the two methods is the interpretation of the coefficients. In linear regression, the coefficients represent the change in the dependent variable associated with a one-unit change in the independent variable, holding all other variables constant. In logistic regression, the coefficients represent the change in the log-odds of the dependent variable associated with a one-unit change in the independent variable, holding all other variables constant.

In summary, linear regression is used when the dependent variable is continuous and the relationship is assumed to be linear, while logistic regression is used when the dependent variable is binary or categorical and the relationship is assumed to be nonlinear. The interpretation of the coefficients is also different between the two methods.

5 Give the Objectives of logistic regression.

Ans.5) The objectives of logistic regression are:

1. **Prediction**: Logistic regression is used to predict the probability of a binary outcome based on one or more predictor variables.

2. **Modeling**: Logistic regression is used to model the relationship between the predictor variables and the probability of the binary outcome.

3. **Parameter estimation**: Logistic regression estimates the parameters of the model, which represent the strength and direction of the relationship between the predictor variables and the probability of the binary outcome.

4. **Hypothesis testing**: Logistic regression can be used to test hypotheses about the relationship between the predictor variables and the probability of the binary outcome.

5. **Model evaluation**: Logistic regression can be used to evaluate the goodness-of-fit of the model and assess the predictive accuracy of the model.

6. **Variable selection**: Logistic regression can be used to select the most important predictor variables in the model.

In summary, the objectives of logistic regression are to predict the probability of a binary outcome, model the relationship between the predictor variables and the probability of the binary outcome, estimate the parameters of the model, test hypotheses, evaluate the model, and select the most important predictor variables.

6 What is Log likelihood?

Ans.6)

- The log likelihood is a measure used in logistic regression to represent the lack of predictive fit.
- It is a function of the parameters of the logistic regression model, which are estimated to maximize the likelihood of the observed data.
- The log likelihood is similar to the sum of squared error in regression analysis, in that it represents the difference between the observed and predicted values of the dependent variable.
- However, unlike the sum of squared error, which is a non-negative quantity, the log likelihood can be negative.
- The log likelihood is used to compare different logistic regression models, with a higher log likelihood indicating a better fit to the data. It is also used in hypothesis testing, to test the significance of the logistic regression coefficients.
- The log likelihood is calculated as the natural logarithm of the likelihood function, which is the probability of observing the data given the parameters of the model.
- The likelihood function is the product of the probabilities of the individual observations, where the probability of each observation is calculated using the logistic regression equation.
- The logistic regression equation is a function of the parameters of the model, which are estimated to maximize the likelihood of the observed data.
- The log likelihood is used to compare different logistic regression models, with a higher log likelihood indicating a better fit to the data. It is also used in hypothesis testing, to test the significance of the logistic regression coefficients.

7 Differentiate between Logistic regression and discriminant analysis.

Ans.7)

Aspect	Logistic Regression	Discriminant Analysis
Dependent Variable	Used when the dependent variable is binary or categorical.	Used when the dependent variable categorical.
Model Type	Generalized linear model.	Not a generalized linear model.
Assumptions	Assumes a nonlinear relationship between the independent variables and the log-odds of the dependent variable.	Assumes a linear relationship betw the independent variables and the dependent variable.
Output	Provides probabilities of class membership.	Provides predicted class membersh directly.
Parameter Estimation	Estimates coefficients using maximum likelihood estimation.	Estimates parameters using various methods like linear discriminant analysis (LDA) or quadratic discriminant analysis (QDA).
Decision Boundary	Uses a threshold to classify observations into different classes.	Uses a linear or quadratic decision boundary to separate classes.
Handling Outliers	Sensitive to outliers due to the logistic function.	More robust to outliers due to the linear or quadratic decision bounda
Sample Size	Suitable for small to large sample sizes.	Suitable for large sample sizes.
Complexity	Simpler to implement and interpret.	More complex due to assumptions about the distribution of the independent variables.

In summary, logistic regression is used for binary or categorical dependent variables, assumes a nonlinear relationship, estimates coefficients using maximum likelihood estimation, provides probabilities of class membership, and is sensitive to outliers. Discriminant analysis, on the other hand, is used for categorical dependent variables, assumes a linear relationship, estimates parameters using methods like LDA or QDA, provides predicted class membership directly, and is more robust to outliers.

8 Explain the Normality of Residual (Error) in both linear and logistic regression.

Ans.8) In both linear regression and logistic regression, the normality of residuals (errors) is an important assumption that affects the validity of the models. Here is an explanation of the normality of residuals in both types of regression:

Linear Regression:

- **Normality of Residuals:** In linear regression, the normality of residuals assumption states that the residuals (the differences between the observed values and the values predicted by the model) should follow a normal distribution.

- **Importance:** Normality of residuals is crucial because it ensures that the statistical inferences made from the model, such as confidence intervals and hypothesis tests, are valid.

- **Checking Normality:** Normality of residuals can be checked visually using a Q-Q plot (Quantile-Quantile plot) or statistically using tests like the Shapiro-Wilk test.

- **Consequences of Violation:** If the normality assumption is violated, it can lead to biased parameter estimates, incorrect standard errors, and inaccurate hypothesis testing results.

Logistic Regression:

- **Normality of Residuals:** In logistic regression, the assumption of normality of residuals does not apply because logistic regression models the relationship between the predictors and the log-odds of the dependent variable, not the dependent variable itself.

- **Residuals in Logistic Regression:** In logistic regression, residuals are not normally distributed because the model does not assume a normal distribution for the dependent variable.

- **Checking Residuals:** Instead of normality, logistic regression focuses on other assumptions like linearity of log-odds and independence of errors.

- **Consequences of Violation:** While normality of residuals is not a concern in logistic regression, violations of other assumptions like linearity or independence can still impact the validity of the model.

In summary, while normality of residuals is a critical assumption in linear regression to ensure the validity of statistical inferences, it is not applicable in logistic regression due to the nature of the model and the way it handles the relationship between predictors and the dependent variable. In logistic regression, other assumptions take precedence, making the normality of residuals less relevant.

9 The logit function(given as l(x)) is the log of odds function. What could be the range of logit function in the domain x = [0,1]?

Ans.9) The logit function, denoted as l(x), is the log-odds function used in logistic regression to model the relationship between the probability of an event occurring and the predictor variables. In the domain x = 1, where x represents probabilities in logistic regression, the range of the logit function l(x) is from negative infinity to positive infinity.

The logit function transforms probabilities (ranging from 0 to 1) into a continuous range that spans from negative infinity to positive infinity. This transformation allows for modeling the linear relationship between the log-odds of the dependent variable and the independent variables in logistic regression.

Therefore, in the domain x = 1, the logit function l(x) can take on any real value, covering the entire range from negative infinity to positive infinity.

OR

The logit function, denoted as l(x), is the log of the odds function used in logistic regression. When the domain of the input variable x is the range 1, the range of the logit function l(x) is from negative infinity to positive infinity.

The logit function is defined as:l(x) = log(x / (1 - x))Where:

- x is the probability value, which can range from 0 to 1.
- l(x) is the logit or log-odds value.

When x = 0, the logit function l(x) = log(0 / 1) = log(0) = -∞ (negative infinity).
When x = 1, the logit function l(x) = log(1 / 0) = log(∞) = +∞ (positive infinity).For any value of x between 0 and 1, the logit function l(x) will take on a value between negative infinity and positive infinity.

The reason the logit function spans this entire range is that it transforms the probability values, which are bounded between 0 and 1, into a continuous scale that can take on any real value. This transformation is crucial in logistic regression, as it allows the model to linearly relate the log-odds of the outcome to the predictor variables.

In summary, when the input variable x is in the domain 1, the range of the logit function l(x) is from negative infinity to positive infinity. This unbounded range of the logit function is a key property that enables logistic regression to model the nonlinear relationship between probabilities and the predictor variables.